In[•]:=
```
(*Redshift Decoupler:Pound-Rebka Asymmetry& Density Lag*)
Manipulate[Module[{t, a, rho, zKin, zGrav, zTotal},
  (*1. Kinematic Scale Factor (Physical Expansion)*)
  (*a(t) represents the physical stretching of space.If kinematicH=0,
  the universe is physically static.*)a[t_] := Exp[kinematicH * t];
  (*2. Retarded Density Gradient*)
  (*The universe was much denser in the past (source).This models the density
   dropping over the photon's travel time.*)rho[t_] := presentDensity +
    (pastDensity - presentDensity) * Exp[-densityDecay * (lookbackMax - t)];
  (*3. Kinematic Redshift Component*)
  (*Standard Doppler/metric stretching effect.*)
  zKin[t_] := a[lookbackMax] / a[lookbackMax - t] - 1;
  (*4. Gravitational Redshift Component (The Asymmetry)*)
  (*The photon loses massive energy climbing out of the dense past,
  but gains very little falling into the sparse present field.*)zGrav[t_] :=
   gravCoupling * Integrate[rho[tau], {tau, lookbackMax - t, lookbackMax}];
  (*5. Total Observed Redshift*)(*The compound observational effect*)
  zTotal[t_] := (1 + zKin[t]) * (1 + zGrav[t]) - 1;
  (*Render the Decoupling Plots*)
  Column[{(*Plot A:Redshift Components*)Plot[{zKin[t], zGrav[t], zTotal[t]},
     {t, 0, lookbackMax}, PlotStyle → {Directive[Thick, Blue, Dashed],
       Directive[Thick, Red, Dotted], Directive[Thick, Purple]}, PlotRange → All,
     Frame → True, FrameLabel → {"Lookback Distance (Billions of Light Years)",
       "Observed Redshift (z)"}, PlotLegends → {"z (Kinematic Expansion)",
       "z (Gravitational Lag)", "z (Total Observed)"}, PlotLabel →
      "Compound Redshift: Decoupling Physical Expansion from Density Lag",
     ImageSize → Large], (*Plot B:The Retarded Gravitational Field*)
    Plot[rho[t], {t, 0, lookbackMax}, PlotStyle → Directive[Thick, Gray],
     PlotRange → {0, pastDensity + 1}, Filling → Axis,
     FillingStyle → Directive[Gray, Opacity[0.2]], Frame → True,
     FrameLabel → {"Lookback Distance (BLY)", "Effective Mass Density of Shells"},
     PlotLabel → "Pound-Rebka Asymmetry: The Density Drop-Off",
     ImageSize → Large]}]], (*Physics Sliders*)
 Style["Physical Expansion (The Standard Model Variable)",
  Bold],
 {{kinematicH, 0.0, "Kinematic Expansion Rate"},
  0.0, 0.2}, Delimiter,
 Style["Gravitational Density Lag (The SO(4,4) Variable)", Bold],
 {{pastDensity, 8.0, "Source Density (Dense Past)"},
  1.0, 15.0},
 {{presentDensity, 0.5, "Receiver Density (Sparse Present)"}, 0.1, 5.0},
 {{densityDecay, 0.3, "Density Decay Rate over Time"}, 0.1, 1.0},
 {{gravCoupling, 0.05, "Gravitational Retardation Coupling"}, 0.0, 0.2},
```

```
    Delimiter,
    Style["Observational Boundary", Bold],
    {{lookbackMax, 13.9, "Optical Boundary Limit (BLY)"}, 5.0, 20.0}]
```

Out[ ]=

Physical Expansion (The Standard Model Variable)

Kinematic Expansion Rate

Gravitational Density Lag (The SO (4,4) Variable)

Source Density (Dense Past)

Receiver Density (Sparse Present)

Density Decay Rate over Time

Gravitational Retardation Coupling

Observational Boundary

Optical Boundary Limit (BLY)



